Complex Engineering Problem

Course Code and Title: EE3004 – Feedback Control Systems
Semester: Spring 2025
Instructors: Dr. Omer Saleem & Ms. Tamania Javaid
Total Marks: 50
Deadline: Monday, 13th May 2025

CLOs and PLOs for Complex Engineering Problem

CLO		Description	Taxonomy Levels	PLO
CLO 5	Theory	Design a closed-loop feedback controller	C5	3

Instructions

Attempt all parts of the following design problem.

Problem Statement

Design a closed loop feedback control system for a customized transfer function $G(s)$. Considering your roll number is $\mathbf{22L-abcd}$, your forward path transfer function would be

$$G(s) = \frac{(s+\mathbf{22})}{(s+\boldsymbol{a})(s+\boldsymbol{b})(s+\boldsymbol{c}+\boldsymbol{d}j)(s+\boldsymbol{c}-\boldsymbol{d}j)}$$

Objectives

1. **Sketch** the root locus assuming it is a unity feedback control system. Show all the calculations related to asymptotes, break points, $j\omega$ crossings and angle of departure/arrival.
2. **Design** a PID controller for your plant such that the maximum overshoot of the overall system would be 12% and the system would have a settling time of 8.0 sec. There should be no error in step response. Show your complete step by step design theoretically.
3. **Design** the circuit of PID compensated system.
4. **Design** a lead-lag compensator using the transient specifications given in part (b) and 15% reduction in steady state error. Show your complete step by step design theoretically.
5. **Design** the circuit of lead-lag compensated system.
6. **Compare** the two controllers in terms of steady state and transient response parameters and suggest their advantages and disadvantages. Also recommend which one should be adopted for implementation.

Deliverables

- Schematic diagrams with detailed annotations and theoretical analysis for the controller design and evaluation of critical gains
- Design Report discussing the design strategy, written in a manner that teaches the theory and step by step application of the theory to solve the above-mentioned problem.
  - ✓ Add appropriate graphs and responses in your report.
  - ✓ Add proper references of the resources you used to learn and solve the problem.
  - ✓ Comment on your method of self-learning, your shortcomings and how would you improve them the next time you try to learn something yourself.

Complex Engineering Problem Attributes

<table>
<tr>
<td rowspan="7">WP1: Depth of knowledge
WP2: Range of conflicting requirements
WP3: Depth of analysis
WP4: Familiarity of issues
WP5: Extent of applicable codes
WP6: Extent of stakeholders
WP7: Interdependence</td>
<td colspan="4">Please fill according to the WPs covered in the course CEP, example is shown here.
• WP1: Depth of Knowledge – Analysis and choice of op-amp principle for each stage requires knowledge of WK3, WK8. Design requires knowledge of WK5, WK8
• WP2: Range of conflicting requirements -- Requirements are conflicting. All objectives are desired, but power consumption should be minimized and DC imperfections must be dealt with. The initial amplification stage design is also critical.
• WP3: Depth of analysis -- No unique solution. Multiple designs possible. Analysis needed to select the most appropriate one.</td>
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<tr>
<td colspan="2">Rubrics</td>
<td>CLOs</td>
<td>Marks</td>
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<tr>
<td>Understands objectives, requirements, and constraints</td>
<td>WP1, WP2</td>
<td>CLO 5</td>
<td>/ 10</td>
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<tr>
<td>Clearly explains the controller design principles for each stage.</td>
<td>WP1</td>
<td>CLO 5</td>
<td>/ 10</td>
</tr>
<tr>
<td>Accurately calculates the controller gains and associated circuit components for the design.</td>
<td>WP1, WP3</td>
<td>CLO 5</td>
<td>/ 10</td>
</tr>
<tr>
<td>Develops a final design that satisfies all requirements</td>
<td>WP1, WP3</td>
<td>CLO 5</td>
<td>/ 10</td>
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<tr>
<td colspan="2">Design Report</td>
<td></td>
<td>/ 10</td>
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<tr>
<td colspan="4">Total</td>
<td>/ 50</td>
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Knowledge Profiles

WK1 - Natural Sciences: A systematic theory-based understanding of natural sciences applicable to the discipline.

WK2 - Mathematics and Computing: The concept-based mathematical thinking, numerical analysis, statistics and formal aspects of computer and information science to support analysis and modelling applicable to the discipline.

WK3 - Engineering Fundamentals: A systematic, theory-based formulation of engineering fundamentals required in an engineering discipline.

WK4 - Engineering Specialization: The knowledge of Engineering specialization that provides theoretical frameworks and bodies of knowledge for the accepted practice areas that are at the forefront in a discipline.

WK5 - Engineering Design: The *Design Thinking Knowledge* that supports engineering design in a practice area of an engineering discipline.

WK6 - Engineering Practice: The Knowledge of engineering practices (technology) in different practice areas of an engineering discipline.

WK7 - Engineering in Society: A systematic, comprehension-based knowledge of the role of engineers in a society and the professional issues related to practicing engineering profession in a discipline: ethics and the professional responsibility of an engineer to public safety including the impact of an engineering activity i.e. economic, social, cultural, environmental and sustainability

WK8 - Research Literature: Engagement with selected knowledge in the research literature of the discipline.

Date: ______________

NAME : Shera Naveed
ROLL NO. : 22L-7651
SECTION : 6C
ASSIGNMENT : CEP (Feedback Control System)

1. Introduction :

Control systems are fundamental to automation of industrial processes in the modern era and play a crucial role in regulating desired outputs for different inputs and disturbances. A closed-loop control system, employs feedback to dynamically correct and adapt its action. In this project, we implement a closed-loop feedback system for a plant characterized by a transfer function from the roll no. 22L-7651. The goal encompasses root locus analysis, control design (PID and lead-lag), and comparative assessment of system performance.

This effort not only investigates classical control theory but also uses stringent mathematical terms to meet transient and steady-state performance requirements. This practical investigation assists in comprehending how theoretical control principles are used in designing real-time stable and effective systems.

2. Problem Analysis :

The transfer function was deduced from the

Date:

roll number 22L-7651, translating to a forward path functions :

$$G(s) = \frac{s+22}{(s+7)(s+6)(s+5+j)(s+5-j)}$$

This is a 4th-order system with a single zero and four poles, including complex conjugate poles. The system's response and stability characteristics are determined by these poles and zeros, making them crucial in design considerations.

To design an effective controller, we need to analyze the root locus of the system to understand how pole locations change with varying gain. This allows determing critical parameters such as asymptotes, break points, jω axis crossings, and angle of arrival or departure.

3. Design Requirements:

The control system must meet specific performance criteria:

- Maximum overshoot : 12%
- Settling time : 8.0 seconds
- Zero steady-state error for a step input

These constraints inform the selection and tuning

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of controllers such as PID and lead-lag. Achieving the desired overshoot and settling time ensures the system's responsiveness and stability, while zero steady-state error ensures accuracy in tracking.

The controller design must also preserve system stability across operating conditions and ensure robust performance despite model uncertainties or external disturbances.

4. Feasibility Analysis :

Given the system's order and pole-zero configuration, classical control techniques like root locus and frequency domain analysis are applicable. These tools are effective in determining system behaviour under proportional gain and aid in placing desired closed-loop poles.

Moreover, the implementation of PID and lead-lag controllers is feasible both in software simulation environments like MATLAB and in hardware circuits using operational amplifiers. This feasibility supports both theoretical and practical validation.

5. Possible solutions :

There are two major approaches to fulfilling the design criteria :

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• PID Controller : A widely used feedback controller that combines proportional, integral, and derivative actions to achieve both transient and steady-state performance.

• Lead Lag Compensator : provides improved phase margin and reduces steady-state error by adjusting the phase and gain characteristics in the frequency domain.

Each method has its strengths. PID control offers ease of implementation and tuning, while lead-lag compensators allow finer control over transient behaviour without excessive gain.

6. Preliminary Design :

Root Locus Analysis :

Our transfer function is :

$$G(s) = \frac{(s+22)}{(s+7)(s+6)(s+5+j)(s+5-j)}$$

the number of poles are : $-7, -6, -5+j, -5-j$

the number of zeros are : -22

Asymptotes :

We have a formula :

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$$= \frac{\Sigma \text{ poles} - \Sigma \text{ zeros}}{\text{no. of pol.} - \text{no. of zeros}}$$

$$= \frac{(-7-6-5-\cancel{j}-5+\cancel{j}) - (-22)}{4-1}$$

$$= \frac{-23+22}{3}$$

$$\boxed{= \frac{-1}{3} \text{ or } -0.333}$$

Angle:

$$\left(\frac{2n+1}{\text{no. of poles} - \text{no. of zeros}}\right)\pi \quad , n = 0, 1, 2, 3$$

for $n = 0 \Rightarrow \boxed{\frac{\pi}{3}}$

for $n = 1 \Rightarrow \boxed{\pi}$

for $n = 2 \Rightarrow \boxed{\frac{5\pi}{3}}$

Breakpoints:

$$\frac{1}{s+22} = \frac{1}{s+7} + \frac{1}{s+6} + \frac{1}{s+5+j} + \frac{1}{s+5-j}$$

$$\frac{1}{s+22} = \frac{s+6+s+7}{(s+7)(s+6)} + \frac{s+5-\cancel{j}+s+5+\cancel{j}}{(s+5)^2-j^2}$$

$$\frac{1}{s+22} = \frac{2s+13}{s^2+13s+42} + \frac{2s+10}{s^2+10s+26}$$

$$= \frac{(2s+13)(s^2+10s+26) + (2s+10)(s^2+13s+42)}{(s^2+13s+42)(s^2+10s+26)}$$

Date: ____________

$$= \frac{2s^3+20s^2+52s+13s^2+130s+338+2s^3+26s^2+84s+10s^2+130s}{s^4+10s^3+26s^2+13s^3+130s^2+338s+42s^2+420s+1092}$$

$$\frac{1}{s+22} = \frac{4s^3+69s^2+396s+758}{s^4+23s^3+198s^2+758s+1092}$$

$$s^4+23s^3+198s^2+758s+1092 = 4s^4+69s^3+396s^2+758s+88s^3+1518s^2+8712s+16676$$

$$s^4+23s^3+198s^2+758s+1092 = 4s^4+157s^3+1914s^2+9470s+16676$$

$$3s^4+134s^3+1716s^2+8712s+15584=0$$

$$\boxed{s=-6.613,\ -27.412,\ -5.32\pm 0.59j}$$

Only $s \simeq -6.61$ lies b/w two real poles $(-7 \,\&\, -6)$, so it is a breakaway point.

jw crossing:

$$\frac{K(s+22)}{(s+6)(s+7)(s+5+j)(s+5-j)}$$

$$= \frac{K(s+22)}{s^4+23s^3+198s^2+(758+K)s+(1092+22K)}$$

making routh table:

s^4	1	198	$1092+22K$
s^3	23	$758+K$	0
s^2	$\frac{3796-K}{23}$	$\frac{22K^2+17768K+827736}{23}$	0

Date:

$s^1 \quad \dfrac{-507K^2-405626K-16160560}{3796-K} \quad 0$

making $s^1 = 0$

$$\frac{-507K^2-405626K-16160560}{3796-K}=0$$

$$-507K^2-405626K-16160560=0$$

$$\boxed{K=-42.05,\ -758}$$

choosing $K=-42.05$

putting K in s^2

$$+\frac{3796-K}{23}s^2+\frac{22K^2+17768K+827736}{23}=0$$

$$\frac{3796+42.05}{23}s^2+\frac{22(-42.05)^2+17768(-42.05)+827736}{23}=0$$

$$166s^2+5195.3=0$$

$$\boxed{s=\pm 5.594}$$

Angle of departure / arrival :

We cannot find angle of arrival as we only have one real zero.

We will find angle of departure from $s_3=-5+j$

we have a formula:

$\theta_d = 180° - (\text{sum of angles from poles to } s_3^*) + (\Sigma \text{ of angles of zeros to } s_3^*)$

Date:

pole at $s_1 = -7$

$$= \tan^{-1}\left(\frac{1}{2}\right) = 26.57$$

pole at $s_2 = -6$

$$= \tan^{-1}\left(\frac{1}{1}\right) = 45°$$

pole at $s_4 = -5-j$

$$= 90°$$

sum of angles from other poles:

$$20.75° + 45° + 90° = 161.57°$$

zero at $s = -22$

$$\tan^{-1}\left(\frac{1}{17}\right) = 3.37$$

putting in formula

$$\theta_d = 180° - 161.57° + 3.37° = 21.8°$$

so, the angle of departure from complex poles $-5 \pm j$ are $\boxed{\pm 21.8°}$

∴ the diagram of this part is in the next ~~st~~ heading.

Date:

7. Design Description:

PID Controller Design:

Given:

$\% \, OS = 12\%$

$T_s = 8 \text{ sec}$

First, we have a formula of dominant pole:

$$s = -\zeta \omega_n \pm j \omega_n \sqrt{1-\zeta^2}$$

Finding ζ:

$$\% \, OS = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100$$

$$0.12 = e^{-\pi\zeta/\sqrt{1-\zeta^2}}$$

$$\cancel{-}2.12 = \cancel{-}\frac{\pi\zeta}{\sqrt{1-\zeta^2}}$$

$$(2.12)^2 = \frac{\pi^2 \zeta^2}{\cancel{\sqrt{1-\zeta^2}}}$$

$$\frac{(2.12)^2}{\pi^2} = \frac{\zeta^2}{1-\zeta^2}$$

$$0.45 = \frac{\zeta^2}{1-\zeta^2}$$

$$0.45 - 0.45\zeta^2 = \zeta^2$$

$$1.45\zeta^2 = 0.45$$

$$\boxed{\zeta = 0.56}$$

Now ω_n:

$$T_s = \frac{4}{\zeta\omega_n}$$

Date:

$$8 = \frac{4}{0.56 \times \omega_n}$$

$$\omega_n = \frac{4}{0.56 \times 8}$$

$$\boxed{= 0.89}$$

putting the values in the formula :

$$s = -(0.56)(0.89) \pm j(0.89)\sqrt{1-(0.56)^2}$$

$$\boxed{s = -0.5 \pm j\,0.73}$$

this is our dominant pole.

PD controller :

Adding zero on the real axis at any point. Now finding the angle of that zero using angle of arrival and departure.

$$\theta_1 = \tan^{-1}\left(\frac{0.73}{22-0.5}\right) = 1.94°$$

$$\theta_2 = \tan^{-1}\left(\frac{0.73}{7-0.5}\right) = 6.40°$$

$$\theta_3 = \tan^{-1}\left(\frac{0.73}{6-0.5}\right) = 7.56°$$

$$\theta_4 = \tan^{-1}\left(\frac{1-0.73}{5-0.5}\right) = 3.43°$$

Date:

$\theta_5 = \tan^{-1}\left(\frac{-1-0.73}{5-0.5}\right) = -21.02$

we have formula

$\theta_z + \theta_1 - \theta_2 - \theta_3 - \theta_4 - \theta_5 = 180$

$\theta_z + 1.94 - 6.40 - 7.56 - 3.43 + 21.02 = 180$

$\theta_z = 174.43^\circ$

$\tan\theta = \frac{P}{B}$

$B = \frac{P}{\tan\theta_z}$

$= \frac{0.73}{\tan(174.43)}$

$B = -7.48 \quad -0.5$

$B = -7.98$

$\boxed{C(s) = s + 7.98} \rightarrow 1$

PI controller:

Now adding a ~~zero~~ pole at origin and a zero at any point on real axis.

Using the values of PD controller:

$\theta_6 = \tan^{-1}\left(\frac{0.73}{0-0.5}\right) = 55.59^\circ$

$\theta_z + \theta_1 - \theta_2 - \theta_3 - \theta_4 - \theta_5 - \theta_6 = 180^\circ$

Date:

$$\theta_2 + 1.94 - 6.40 - 7.56 - 3.43 + 21.02 + 55.59 = 180°$$

$$\theta_2 = 118.84°$$

$$\tan\theta = \frac{P}{B}$$

$$B = \frac{P}{\tan\theta}$$

$$= \frac{0.73}{\tan(118.84)}$$

$$B = -0.401 - 0.5$$

$$= -0.901$$

$$\boxed{C(s) = \frac{s + 0.901}{s}} \rightarrow 2$$

PID Controller :

$$K\,G(s)H(s) = -1$$

$$|K\,G(s)H(s)| = |-1|$$

$$K = \left|\frac{1}{G(s_0)H(s_0)}\right|$$

$$H(s) = 1$$

$$G(s) = \frac{s+22}{(s+7)(s+6)(s+5+j)(s+5-j)}$$

$$K = \frac{(s_0+7)(s_0+6)(s_0+5+j)(s_0+5-j)}{s_0+22}$$

Date:

a) $s_0 + 22 = -0.5 + 0.73j + 22 = 21.5 + 0.73j = 21.51$

b) $s_0 + 7 = -0.5 + 0.73j + 7 = 6.5 + 0.73j = 6.54$

c) $s_0 + 6 = -0.5 + 0.73j + 6 = 5.5 + 0.73j = 5.54$

d) $s + 5 + j = -0.5 + 0.73j + 5 + j = 4.5 + 1.73j = 4.82$

e) $s + 5 - j = -0.5 + 0.73 + 5 - j = 4.5 - 0.27j = 4.50$

$$K = \frac{(6.54)(5.54)(4.82)(4.50)}{21.51}$$

$$\boxed{K = 36.53} \rightarrow 3$$

Multiplying 1 2 & 3

$$36.53\frac{(s + 7.98)(s + 0.901)}{s}$$

$$= 36.53\frac{(s^2 + 0.901s + 7.98s + 7.18)}{s}$$

$$= \frac{36.53s^2 + 324.4s + 262.28}{s}$$

$$= 36.53s + 324.4 + \frac{262.28}{s}$$

$$324.4 = \frac{R_2}{R_1} + \frac{C_1}{C_2} \rightarrow 1$$

$$36.53 = R_2 C_1$$

$$262.28 = \frac{1}{R_1 C_2}$$

Assuming $\boxed{R_2 = 1K\Omega}$

Date:

$$36.53 = R_2 C_1$$

$$C_1 = \frac{36.53}{1K}$$

$$\boxed{C_1 = 36.53mF}$$

$$262.28 = \frac{1}{R_1 C_2}$$

$$R_1 C_2 = \frac{1}{262.28}$$

$$C_2 = \frac{1}{262.28 R_1}$$

$$324.4 = \frac{R_2}{R_1} + \frac{C_1}{C_2}$$

$$324.4 = \frac{1000}{R_1} + 36.53m \times 262.28 R_1$$

$$324.4 R_1 = 1000 + 9.581 R_1^2$$

$$9.581 R_1^2 - 324.4 R_1 + 1000 = 0$$

$$R_1 = 3.43\,\Omega, \quad \boxed{R_1 = 30.42\,\Omega.}$$

choosing $R_1 = 30.42\,\Omega$

$$C_2 = \frac{1}{262.28(30.42)}$$

$$\boxed{C_2 = 125.33\mu F}$$

Date:

Lead Lag Compensator Design

Lead compensator:

First add a zero then add a pole any where in the real axis.

Now finding the angle of zero and pole.

$$\theta_{zc} - \theta_{pc} + \theta_1 - \theta_2 - \theta_3 - \theta_4 + \theta_5 = 180°$$

$$\theta_{zc} - \theta_{pc} + 1.94 - 6.40 - 7.56 - 3.43 + 21.02 = 180°$$

$$\theta_{zc} - \theta_{pc} = 174.43°$$

Assuming $P_c = -25$

$$\theta_{pc} = \tan^{-1}\left(\frac{0.73}{25 - 0.5}\right)$$

$$\theta_{pc} = 1.706$$

$$\theta_{zc} = 174.43 + 1.706$$

$$\theta_{zc} = 176.026$$

$$b = \frac{P}{\tan(176.026)}$$

$$b = -10.5 - 0.5$$

$$b = -11$$

the location of pole is -25 and location of zero is -11.

Date:

$$C(s) = \frac{s + z_c}{s + p_c}$$

$$C(s) = \frac{s+11}{s+25} \rightarrow (1)$$

Lag Compensator:

$$e_{ss} = \frac{1}{1+K_{p_o}}$$

$$K_{p_o} = \lim_{s\to 0} G(s)$$

$$= \frac{22}{7\times 6 \times 5+j \times 5-j}$$

$$= \frac{22}{42\times 26}$$

$$K_{p_o} = 0.02$$

$$ess = \frac{1}{1+0.02}$$

$ess = 0.98$

its 15% is 0.147

Removing 15%, the error is

$e_{ss} = 0.833$

$$ess = \frac{1}{1+K_{p_c}}$$

Date:

$$1 + K_{p_c} = \frac{1}{0.833}$$

$$1 + K_{p_c} = 1.20$$

$$K_{p_c} = 0.20$$

Lead Lag Compensator:

the formula is

$$K_{p_c} = K_{p_0}\left(\frac{z_c}{p_c}\right)$$

Assuming $p_c = -0.01$

$$0.20 = \left(\frac{0.02}{0.01}\right) z_c$$

$$z_c = -0.1$$

the equation becomes

$$\frac{s+0.1}{s+0.01} \rightarrow ② \Rightarrow \frac{s - z_c}{s - p_c}$$

multiplying ① & ②

$$= \left(\frac{s+11}{s+25}\right)\left(\frac{s+0.1}{s+0.01}\right)$$

$$= \frac{s^2 + 0.1s + 11s + 1.1}{s^2 + 0.01s + 25s + 0.25}$$

$$= \frac{s^2 + 11.1s + 1.1}{s^2 + 25.01s + 0.25} \rightarrow Ⓐ$$

Date:

the numerator of the equation which we will use to find the values of R_1, C_1, R_2 and C_2 is

$$\left(s+\frac{1}{R_1C_1}\right)\left(s+\frac{1}{R_2C_2}\right)$$

$$= s^2+\left(\frac{1}{R_1C_1}+\frac{1}{R_2C_2}\right)s+\frac{1}{R_1C_1R_2C_2}$$

$$= s^2+\frac{R_1C_1+R_2C_2}{R_1C_1R_2C_2}s+\frac{1}{R_1C_1R_2C_2} \rightarrow \text{num}$$

and the denominator is:

$$s^2+\left(\frac{1}{R_1C_1}+\frac{1}{R_2C_2}+\frac{1}{R_2C_1}\right)\cdot s+\frac{1}{R_1R_2C_1C_2} \rightarrow \text{den}$$

comparing eq (1) numerator and num,

$$\frac{R_1C_1+R_2C_2}{R_1C_1R_2C_2} = 11.1$$

$$R_1C_1+R_2C_2 = 11.1R_1C_1R_2C_2 \rightarrow (1)$$

$$\frac{1}{R_1C_1R_2C_2} = 1.1$$

$$R_1C_1R_2C_2 = 0.90 \rightarrow (2)$$

put eq (2) in eq (1)

Date:

$$R_1C_1 + R_2C_2 = 11.1 \times 0.90$$

$$R_1C_1 + R_2C_2 = 9.99 \rightarrow ③$$

comparing denominator of eq Ⓐ with den

$$\frac{1}{R_1C_1} + \frac{1}{R_2C_2} + \frac{1}{R_2C_1} = 25.01$$

$$\frac{R_2^2C_1C_2 + R_1R_2C_1^2 + R_1R_2C_1C_2}{R_1R_2^2C_1^2C_2} = 25.01$$

$$\frac{\cancel{C_1R_2}(R_2C_2 + C_1R_1 + R_1C_2)}{\cancel{C_1R_2}(R_1R_2C_1C_2)} = 25.01$$

$$C_1R_1 + C_2R_2 + R_1C_2 = 25.01\, R_1R_2C_1C_2$$

put value of eq ②

$$C_1R_1 + C_2R_2 + C_2R_1 = 25.01 \times 0.90$$

put eq ③ in this equation

$$9.99 + C_2R_1 = 22.51$$

$$C_2R_1 = 12.519$$

Assuming $R_1 = 1K\Omega$ and $C_1 = 1\mu F$

$$C_2 = \frac{12.519}{1K}$$

$$C_2 = 12.51mF$$

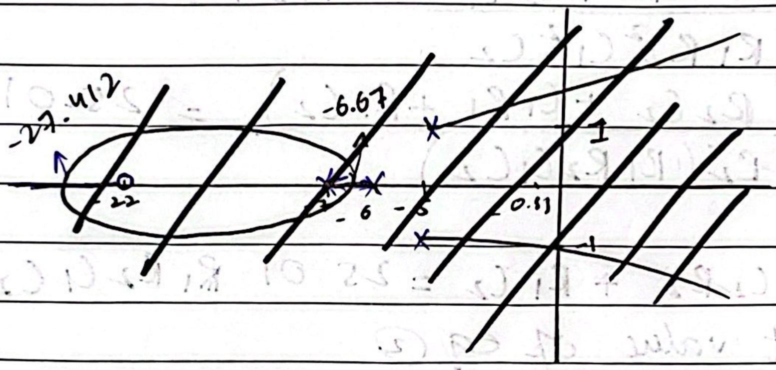
Date:

put all the values in eq. 2

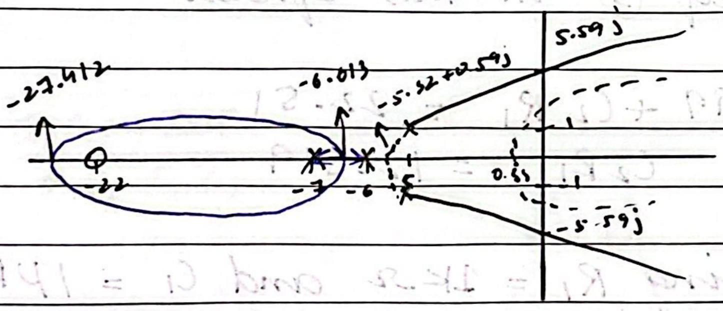
$$1000 \times 1\mu \times 12.51\,m \times R_2 = 0.90$$

$$R_2 = 71.8\ K\Omega$$

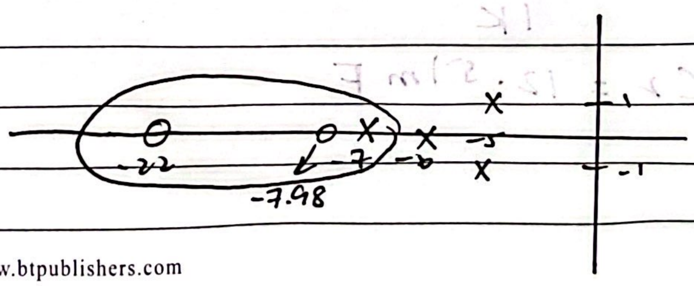
8. **Experimental Results :**



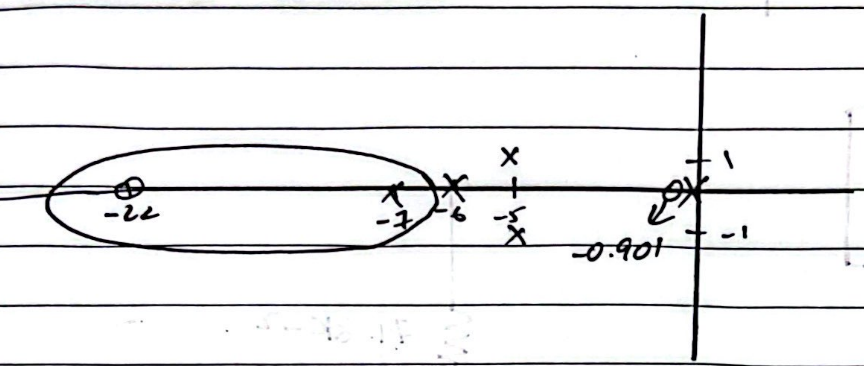
the root locus diagram is



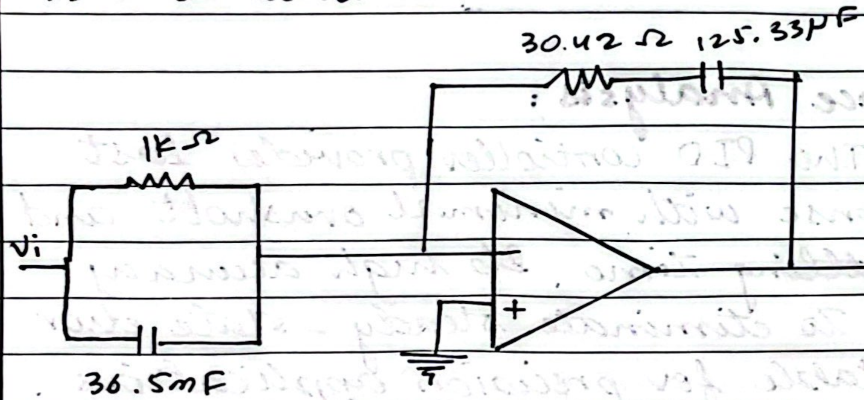
For PD controller



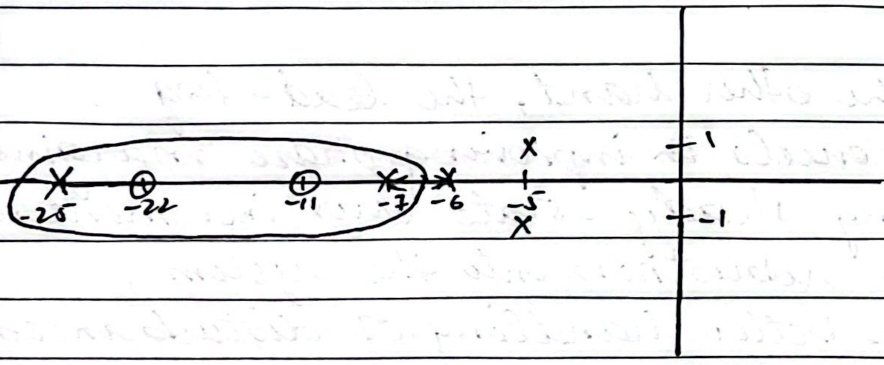
For PI controller :



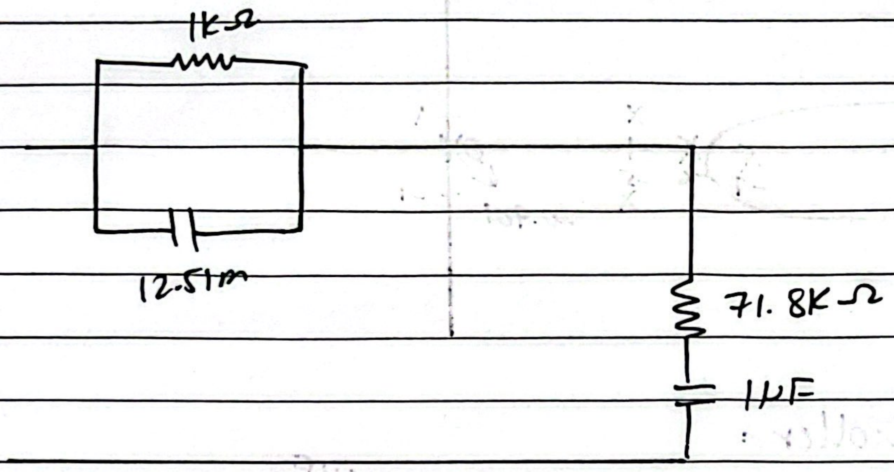
PID controller :



For lead compensator :



Lead Lag compensator :



9. Perfamance Analysis :

The PID controller provides fast system response with minimal overshoot and acceptable settling time. Its high accuracy and ability to eliminate steady-state error makes it suitable for precision applications. However, it is more sensitive to sensor noise and requires proper filtering and derivative term adjustment.

On the other hand, the lead-lag compensator excels in improving phase margins and reducing steady-state error incrementally. It introduces robustness into the system, allowing for better handling of disturbances and parameter changes. However, the system response is generally slower compared to a PIDcontrolled system.

From a control engineering standpoint, PID control is more straightforward for real-time applications, while lead-lag compensators are favoured in systems requiring higher robustness and refined error control.

10. Future scope:

Future extensions of this project can explore digital control methods using micro controllers or DSP. This would allow implementation of adaptive control systems dynamics.

Furthermore, techniques like fuzzy logic control or model predictive control (MPC) can be integrated to handle non-linearities and time delays. Combining classical control methods with modern AI techniques could lead to more intelligent and autonomous systems.

11. Social and cultural Implication:

The widespread application of control systems in sectors such as transportation, manufacturing, energy, and healthcare implies significant social impact. For instance, precise control in medical devices improves patient outcomes,

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while automation in agriculture enhances productivity and reduces manual labor.

Culturally, the advancement and application of control system contribute to technological litrary and innovation in society. As these systems become embedded in everyday technologies, they empower communities to operate smarter, more efficient, and safer systems, directly contributing to quality of life improvements.

12. Conclusion:

A complete closed-loop control system was designed using classical techniques for a 4th order plant. Through rigorous root locus analysis, jω-axis evaluations, and angle calculations, both PID and lead lag controllers were derived.

The comparative analysis concluded that while the PID controller offers rapid response and simple implementation, the lead-lag compensator provides superior steady-state performance and robustness. Ultimately, the choice dependes on the system's operational demands. This exercise underscores the relevence

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of the control theory in engineering practice and its broader implications across industries and society